

A MODEL CONFIGURATIONS

Table 1: Default model configurations across all architectures. In experiments, we sweep learning rate and embedding dimension, reporting results from the instance with highest accuracy.

(a) Attention		(b) Hyena		(c) BaseConv	
Parameter	Values	Parameter	Values	Parameter	Values
dropout	0.0	l_max	1024	l_max	1024
num_heads	1	filter_order	64	kernel_size	[3, -1]
		num_heads	1	implicit_long_conv	True
		num_blocks	1	use_act	False
		outer_mixing	False		
		dropout	0.0		
		filter_dropout	0.0		
		short_filter_order	3		
		bidirectional	False		
(d) Based		(e) H3		(f) Mamba	
Parameter	Values	Parameter	Values	Parameter	Values
<i>BaseConv Layer</i>		l_max	1024	d_conv	4
l_max	1024	d_state	1024		
kernel_size	3	head_dim	1024		
implicit_long_conv	True				
use_act	False				
<i>Based Layer</i>					
l_max	1024				
feature_dim	8				
num_key_value_heads	1				
num_heads	1				
feature_name	taylor_exp				
train_view	quadratic				

Param.	Description
H	Is the head terminal at the left or the right of each production?
$d_{\max}$	Maximum depth permitted for the PCFG to generate.
$L_{\max}$	Maximum number of symbols of the right-hand side of a production rule.
$R_{\max}$	Maximum number of production rules for each nonterminal.
$ \mathcal{N} $	Number of nonterminal symbols in the PCFG vocabulary.
$ \Sigma $	Number of terminal symbols in the PCFG vocabulary.
r_{Σ}	Relative weightage on choosing a terminal when sampling production rules.

Table 2: Parameters used for constructing a PCFG. We define PCFGs in Greibach Normal Form (GNF); see Appendix B for more details.

B FORMALISATION OF ATR

For reference, we provide formal definitions for PCFGs and the normal form we use in ATR.¹⁰

Definition B.1. A **probabilistic context-free grammar** is a tuple $\mathcal{G} = \langle \mathcal{N}, \Sigma, S, \mathcal{R}, p \rangle$ where:

- $\mathcal{N}$ is a finite set of non-terminal symbols;
- Σ is an alphabet of terminal symbols;
- $S \in \mathcal{N}$ is a start symbol;
- $\mathcal{R} \subset \mathcal{N} \times (\mathcal{N} \cup \Sigma)^*$ is a finite set of production rules, mapping a left-hand side symbol $N \in \mathcal{N}$ to a string of symbols that may be either terminals or nonterminals; each such rule is written as $X \rightarrow \alpha$;
- $p : \mathcal{R} \rightarrow [0, 1]$ is a weighting function which assigns a probability to each production rule for a nonterminal; this function is locally normalised, meaning $\{\sum_{X \rightarrow \alpha} p(X \rightarrow \alpha) = 1 \mid X \in \mathcal{N}\}$.

Definition B.2. A PCFG $\mathcal{G} = \langle \mathcal{N}, \Sigma, S, \mathcal{R}, p \rangle$ is in **Greibach normal form (GNF)** if each production rule in $\mathcal{R}$ is of the form $X \rightarrow a X_1 \dots X_n$, where $X_1, \dots, X_n \in \mathcal{N}$ and n may be 0. Similarly, a PCFG is in **right-Greibach normal form** if each rule is of the form $X \rightarrow X_1 \dots X_n a$.

For ATR, the PCFG is in Greibach normal form if the head is the leftmost symbol of the production rule’s righthand side; similarly, if the PCFG is right-headed, it is in right-Greibach normal form.

Definition B.3. A **derivation step** $\alpha \Rightarrow \beta$ is an operation where, given strings of symbols $\alpha, \beta \in (\mathcal{N} \cup \Sigma)^*$, the leftmost nonterminal $X \in \mathcal{N}$ in α is rewritten using the right-hand side of a production rule $X \rightarrow \dots \in \mathcal{R}$ to obtain β .

Definition B.4. A **derivation** under the PCFG $\mathcal{G}$ is a sequence of strings $[\alpha_0, \dots, \alpha_m]$ where $\alpha_0 \in \mathcal{N}$ and each step α_{i+1} is formed by a derivation step on α_i . The final string $\alpha_m \in \Sigma^*$ is the **yield** of the derivation.

Each ATR document is the yield of a derivation sampled under the GNF PCFG $\mathcal{G}$.

B.1 ADDITIONAL DETAILS ON ATR

Parent terminals in GNF. We set the left/right-most terminal in each production rule (which leads to the GNF property) the parent of all other generated terminals. This terminal is sampled specially: for each nonterminal, we independently sample a distribution over terminals from a uniform Dirichlet, and for all production rules with that nonterminal on the lefthand side we use that distribution to sample the parent terminal. This simulates how heads of phrases in natural language (analogous to our parent terminals) decide the type of the phrase they head (analogous to our nonterminals).

Maximum depth. To enforce maximum depth, we first assign a uniformly random depth score $d : \mathcal{N} \rightarrow \mathbb{N} \in \{1, \dots, \max_depth\}$ to each nonterminal in the vocabulary. Then, for each production rule for each nonterminal X , we only allow nonterminals Y with $d(Y) > d(X)$ on the right-hand side. Note that this means no recursion is possible.

¹⁰We use similar formalisations of PCFGs as previous work in NLP, e.g. Nowak & Cotterell (2023).

C TASK HYPERPARAMETERS

Table 3: Task hyperparameters used for constructing key–value sets for AR and PCFGs for ATR. For a description of each parameter, see Table 2.

(a) Parameters used for constructing AR documents.

Parameter	Values
$L_{\max}$	32
$L_{\min}$	32
$ \Sigma $	{8192}

(b) Parameters used for constructing ATR documents.

Parameter	Values
H	Right
$d_{\max}$	10
$L_{\max}$	5
$R_{\max}$	5
$ \mathcal{N} $	40
$ \Sigma $	20
r_{Σ}	20